NailBender: Exploring Nail Flexion for Haptic Input and Output

Yanjun Chen
University of Illinois Urbana-Champaign
Champaign, Illinois, USA
yanjunc2@illinois.edu

Xiping Sun
University of Pennsylvania, Grasp Lab
Philadelphia, Pennsylvania, USA
sxp22@seas.upenn.edu

Jung-Hwan Youn
University of Illinois Urbana-Champaign
Urbana, Illinois, USA
jy92@illinois.edu

Craig Shultz
University of Illinois Urbana-Champaign
Urbana, Illinois, USA
shultz88@illinois.edu

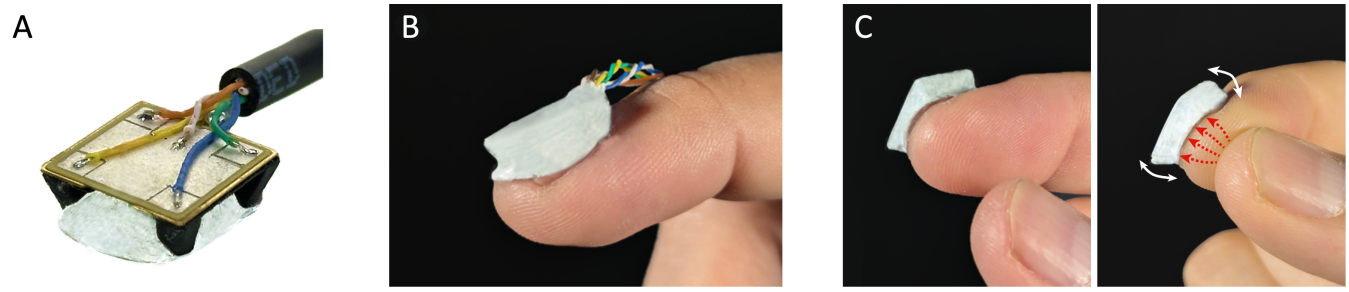


Figure 1: A) NailBender prototype device. B) Overview of NailBender adhered to a user's finger, covered in soft silicone. C) NailBender devices leave the user's fingerpad totally unobstructed, but can still detect applied forces and render output effects, all through the bending of the user's fingernail.

Abstract

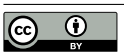
We introduce NailBender, a lightweight (<1 g) nail-mounted device that uses a single piezoelectric plate with segmented electrodes for both sensing and actuation. At ±90 V, the piezo produces 50 µm of deflection across 1–100 Hz, requiring just 40 mW at 20 Hz. Actuation remains detectable down to ±20 V. The piezo can also sense forces from 0.1 to 24 N applied to the bare fingerpad, as well as heartbeat and environmentally coupled vibrations. Using NailBender, we demonstrate single and multiple input/output configurations: variable-force pinch buttons (3 levels, 90.7% accuracy), spatially differentiated left/right tactile actuation (5 patterns, 83.2% recognition), and ML-based micro-gesture classification (5 gestures, 98% accuracy after per-user calibration). These results show that NailBender can enable expressive, bidirectional haptic interaction in a compact, wearable form factor without blocking the fingerpad.

CCS Concepts

• **Human-centered computing** → **Haptic devices**.

Keywords

Fingernail wearable, piezoelectric, sensing and actuation, haptic feedback, tactile sensing



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1 Introduction

Our hands are essential to how we interact with the world, affording us the ability for fine manipulation, expressive gestures, and precise tactile sensing. As computing moves toward mixed and augmented reality, there is a growing need for complex input and output paradigms that support the nuanced natural abilities of our hands without requiring bulky controllers or gloves. Optical hand tracking supports expressive touch input, and it, along with gaze, is becoming a standard in MR interactions [55]. Precise finger contact and force detection, however, is nontrivial, especially beyond a single degree of freedom [48], and haptic feedback is largely limited to self or environmentally referenced passive contact [76].

Haptic feedback enabled fingertip-mounted wearables, which we term "finger-mounted displays" (FMDs), offer a compelling research direction to fill this gap. Of special interest are FMDs which do not block the user's natural tactile dexterity or perception (so-called tactually transparent, or augmented reality haptic displays [75, 79, 83, 87]). We see low-encumbrance FMDs as being crucial to explore and develop if generalized haptic feedback is to be a part of the future of augmented and mixed reality computing. Specifically, we propose that nail-based FMDs hold promise for both enhancing

sensing fidelity and providing rich, spatially coherent tactile feedback in touch, pinch, and force based spatial interactions, aspects which ring based or wrist based wearables have difficulty augmenting due to their spatial separation from the point of interaction.

To achieve this promise, nail-based FMDs have a number of challenges to overcome. Core challenges in the wearable haptic display design space include: device compactness, low weight, minimal motion impairment, and good long-term comfort [52], while more recent reviews of the area highlight device mechanical coupling and grounding, customization and fitting, and enhanced stimuli richness [18]. When specifically looking at nail-based devices, a recent work has explicitly called for future work to focus on "integration into artificial nails", "scaling to other sensing techniques", "coupling force input", "scaling to multiple actuators", and developing a new "actuation mechanism" to meet the challenges of the space [76]. When considering all of these different factors as a whole, we asked the following research question: "What might be an underutilized physical signal that offers rich sensing and actuation opportunities while still satisfying device wearability concerns?"

The answer we explore in this work is nail flexion, or the spatial bending of the fingernail. We propose that this overlooked biomechanical signal can encode rich information about interaction, as our nails bend constantly throughout the day in response to interaction forces as our hands manipulate the world around us [4, 39, 59]. Furthermore, our perceptual system appears to be highly attuned to nail flexion signals, and nail-mediated tactile signals have recently been shown to have sensitivity that can, in some cases, rival the sensitivity of the fingerpad [39]. To investigate nail flexion in greater depth, we developed NailBender, a first of its kind, piezo-based, nail-mounted haptic platform that supports both input and output interactions.

The novelty of NailBender is not in the piezoelectric material itself but in demonstrating that a compact, nail-mounted piezoelectric platform can simultaneously support high-sensitivity, multi-dimensional input and expressive multidimensional haptic output without obstructing natural touch. Placement and coupling are core technical barriers in wearable haptics [18, 52, 76], and our results show that mounting a piezo on the nail enables a mode of rich, indirect interaction not previously demonstrated. This dual capability addresses needs repeatedly raised in prior fingertip haptics work for less bulky and better coupled devices, showing that a single, well-coupled transducer can serve both directions on an underutilized and unobtrusive channel.

The remainder of this work proceeds in three parts. We begin by detailing the working principles of nail flexion and the design of our device. We then present a set of technical characterizations and user studies that probe NailBender's various configurations: single input and single output, multi-output, and multi-input. In each of these cases we highlight the NailBender platform's ability to render sensations, sense forces, and support interactive tasks. We developed illustrative examples based on application scenarios from prior work (mediation of information, on-body AR user-interfaces and augmenting passive props) to provide context for NailBender's capabilities, and close with a discussion of limitations and directions for future work. Through this exploration, we argue that nail flexion offers a compelling yet largely unexplored channel for fingertip wearables and mixed-reality interaction.

2 Related Work

2.1 Micro-gesture Hand/Finger Sensing

Micro-gestures have been increasingly explored as a natural input strategy, mapped to controllers for everyday interactions [24, 63] as well as intuitive interfaces in AR/VR environments [29, 38]. They are finger-scale motions commonly categorized as free hand [6], self-haptic [68], and object based interactions [64, 84]. Self-haptic micro-gestures such as thumb to finger interactions [8] exploit the hand's fine motor skills to enable expressive, efficient and unobtrusive input. To characterize this interaction space, prior work has proposed design spaces for single-hand thumb to finger micro-gestures, outlining gesture primitives such as taps, pinches, slides, draws, and flick [15], and exploring their combinations for richer input [68]. These fine-grained gestures inevitably introduce sensing challenges, mainly the need for high-resolution detection and the burden of wearable instrumentation.

Extensive research has explored the detection of finger input through two main directions: vision-based systems [47, 49] and sensor-based approaches [73]. Vision-based gesture recognition has been extensively studied [57], from earlier image-based methods [85] to depth-camera pose estimation [58] and more recent deep learning approaches [7, 17, 32, 50]. Kin et al. [29] proposed a lightweight TCN model that has been applied to real-time micro-gesture recognition in AR/VR systems. Nevertheless, challenges remain with background noise, illumination, real-time performance, and the recognition of fine-grained micro-gestures.

Sensor-based approaches leverage diverse physical signals to capture micro-gestures with high precision. These include optical methods such as acoustic–optic [86] and infrared sensing [19, 44], mechanical methods such as pressure [14], barometric [67], ultrasound/acoustic sensing [23, 31, 90], and physiological or field-based methods such as IMU [36, 77], EMG [25, 45, 60], magnetic [10, 53], electric field [11, 27], impedance tomography [91], and radar [35, 37]. Many of these modalities have been implemented in wearable form factors, Kaifosh et al. [25] proposed a wrist-worn sEMG interface for fine-grained micro-gestures such as tapping, swiping, and pinching. Beyond wrist-worn devices, another representative form is ring-based sensors worn close to the fingers, which capture subtle inputs [28, 80, 81]. MouseRing [66] demonstrates this approach with continuous finger sliding on surfaces. These devices highlight common design considerations around wearability, freedom of hand movement, and fine-grained sensing.

2.2 Fingertip Wearables

The human finger is a natural medium for interacting with physical objects. Accordingly, a range of research has focused on developing fingertip wearables that deliver haptic feedback. These devices vary in form factor and interaction objectives, ranging from glove-based systems to compact fingertip interfaces. Haptic gloves, in particular, can provide diverse haptic feedback, such as vibrotactile or kinesthetic cues [5, 82]. Several commercialized haptic gloves, including the HaptX Glove G1 [20], the TactGlove DK2, and the SenseGlove Nova 2 [1], combine finger-motion tracking with multiple haptic actuation mechanisms to enable immersive VR interactions.

Fingertip-mounted devices, on the other hand, feature a compact form factor that minimizes the burden on fingertip movement.

Various such devices have been introduced in HCI, delivering vibrotactile or force feedback directly to the fingerpad using small DC motors or vibration actuators [46, 61, 75, 89]. While promising for VR interaction, they still face challenges in achieving a more compact and lightweight design, largely due to the size and weight of the motors. To address this, researchers have explored novel actuation mechanisms to build compact, lightweight fingertip-mounted devices capable of delivering localized interactions [65, 88].

With the rise of mixed reality (MR), which requires interaction with both real and virtual environments, researchers have focused on tactually transparent haptic devices. Thin-film electrotactile devices [83] and perforated electrotactile designs with extra holes [74] have been developed to provide effective tactile feedback while minimizing encumbrance. Other approaches avoid blocking the fingerpad entirely by targeting alternative regions, such as the edges of the fingertip [41, 72, 87] or palmar side of the hand [71].

2.3 Nail Based Devices

Most finger-mounted displays (FMDs) use ring or thimble form factors. However, rings offer limited sensing of fingerpad forces, while thimble devices occlude the pad and interfere with natural manipulation and touchscreen use. As an alternative, the nail's rigid structure makes it an attractive mounting site for fingertip wearables, and nail-based devices have emerged as a promising platform for tactually transparent gesture sensing and haptic feedback without obstructing the fingerpad. We summarize some of the most closely related works in this area in Table 1.

Concerning touch input, there are a number of approaches. Researchers have turned the nail surface into a touch-sensitive interface using miniature capacitive sensors [26, 33, 51]. Although straightforward, this approach requires contact on the nail surface, which is ergonomically awkward and limits the range of gestures. Other techniques infer fingerpad interaction from signals around the nail, such as optical sensing of blood volume beneath the nail for estimating normal and shear fingertip forces and posture[42], or asymmetric nail deformation using a micro cameras [34], or multiple strain gauges [12, 22, 59, 78]. More recently, active excitation of the fingernail has enabled sensing of contact force and edge orientation without covering the fingerpad. [30]. From our review, no prior work has used piezos for nail based strain sensing.

Turning to tactile output, researchers primarily utilized vibrotactile feedback. Early work mounted single compact vibrational motor on the nail to render virtual textures during natural fingerpad contact [2]. Building on this concept, NailTactors [21] mounted an array of vibrational motors on the nail to convey both static and dynamic spatial cues. Despite reduced compactness and weight added from multiple motors, the system demonstrated promising eyes-free interaction by delivering varied cues to the user. Haptlets also showed a single wide-bandwidth LRA with wireless communication and battery setup [56]. A single report of piezo-based haptics has been reported [69], though the vibration was limited to peak tactile sensitivity (200-300 Hz), and not psychophysically validated.

Only a select few systems have attempted both tactile sensing and feedback, typically using separate devices for each. Some nail-based devices have explored closed-loop interaction using additional external sensors, such as on adjacent skin regions [43, 76], but this can limit applicability across diverse environments. vARitouch demonstrated a nail-worn device with both sensing and feedback, rendering compliance for rigid objects [79]. However, these nail-based devices measure only a single contact pressure and provide only a single vibrational output. This line of work also highlights several open challenges, including integration into artificial nails, input–output coupling, and scaling to multiple actuators and sensing channels. NailBender opens up this underexplored region of the design space by unifying rich input sensing and multi-output haptics in a single lightweight, nail sized form factor.

Table 1: Nail-based devices by form factor and tactile I/O.

Ref	Form Factor		Input		Output (Haptic)	
	Size	Weight	Modality	M	Modality	M
FingerPad [9]	Small	Light ($* < 1$ g)	Magnet sensor	✓	-	-
Nail+ [22]	Small	Light ($* < 0.1$ g)	Strain gauges	✓	-	-
Strain2Force [12]	Small	Light (0.3 g)	Strain gauges	✓	-	-
Nail-tip [51]	Small	Light ($* < 0.1$ g)	Capacitive sensors+ Strain gauge	✓	-	-
Sakuma'18 [59]	Small	Light ($* < 1$ g)	Strain gauges+ Accelerometer	✓	-	-
PZT-MEMS [69]	Small	Light (0.05 g)	-	-	Piezo	×
Haplets [56]	Small	Heavy (5.2 g)	-	-	LRA	×
Fingeret [41]	Large	Heavy (18 g)	-	-	DC motor+LRA	✓
NailTactors [21]	Large	Heavy ($* > 4$ g)	-	-	Multiple ERMs	✓
vARitouch [79]	Large	Medium (*2 g)	Photo-reflective sensor	×	LRA	×
This work	Small	Light (0.84 g)	Segmented Piezo	✓	Segmented Piezo	✓

M: multimodal sensing or haptic channels. *Approximation based on published data.

3 Working Principles

The design of NailBender is grounded in two main insights. First, fingernails play a critical, yet overlooked, role in tactile perception, dexterity, and object manipulation. Early work on a tactile nail chip further showed that mechanically deforming the nail can modulate tactile perception without covering the fingerpad [70]. The nail bed is richly innervated and contributes to force encoding and spatial acuity. This means nails are not just passive protective structures but active participants in how we sense and control fine forces. Leveraging this opens opportunities for nuanced sensing and richer haptic output.

Second, the fingernail offers a uniquely stable and socially acceptable site for mounting sensors and actuators. Its rigid structure and proximity to the fingertip enable tight mechanical coupling and efficient force transfer. Alternatives such as rings or wristbands are farther from the point of contact and provide weaker, less consistent coupling. These qualities position the nail as both a perceptual substrate and a practical physical interface for HCI.

3.1 Nail Flexion Encodes Tactile Information

Fingernails are not simply passive structures that protect our fingertips. They play an active role in tactile perception and manual dexterity [39], and nail loss or damage, through injury or disease,

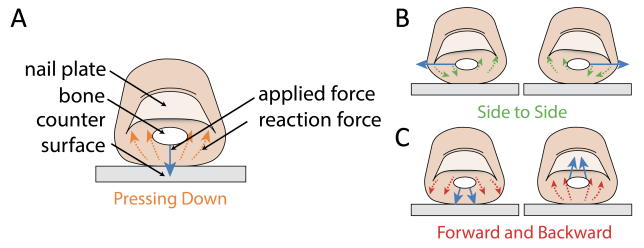

Figure 2: A) Overview of finger forces and biomechanics under fingerpad loading. B) Loading under side to side motion. C) Loading under forward and backward motion.

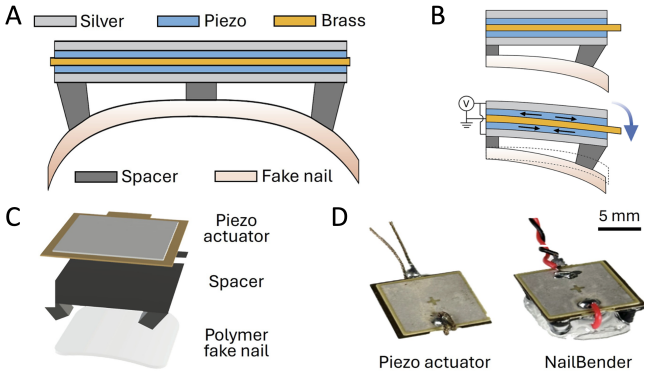

Figure 3: A) Stackup schematic of NailBender. B) Piezo bending mechanism under applied voltage. C) Stackup with rendered 3D model. D) Core actuator and assembled NailBender.

has been linked to impaired manual dexterity [3, 16, 40, 92]. Furthermore, humans can localize haptic stimuli on the nail with nearly the same acuity as the fingerpad [39, 62], as the skin beneath the nail (the nail bed) is innervated with a rich set of mechanoreceptors. Additionally, receptors around the nail edges encode forces from the fingerpad [4], and are more sensitive to high forces than receptors in the middle of the fingerpad [87]. Little is known about precisely how this encoding occurs, although it has been investigated for activity detection in HCI contexts [22, 59].

Presumably, forces from the fingertip are coupled to the fingernail through the bulk of the finger and around the bone, which anchors the nail through a fibrous network of tissue. Applied force is generated by the muscles in the forearm, which pull on tendons connected to the finger bone. This creates a downward force (Figure 2A) that is opposed by the contact surface, generating internal pressure around the bone and against the nail plate. This reaction force causes upward deflection at the nail edges, while the center remains tightly anchored to the bone. Forces applied side to side (Figure 2B) or forward and backward (Figure 2C) create differential strains in the nail plate [59]. NailBender can both sense and actuate similar strains through an external structure attached to the nail.

3.2 Interface for Actuation and Sensing

Nails provide a stable mounting point and are socially acceptable for wear, provided the fake nail remains low-profile. Our NailBender prototype weighs under 1 g, with the piezo bender plate itself only 0.4 mm thick and weighing 0.5 g. Directly gluing such a plate to the nail would only produce high-frequency vibration, similar to [69], since piezo devices are strain-limited and, when mounted beneath stiff surfaces like touchscreens, are used almost exclusively for vibrotactile output. To overcome this, we attach a thin fake nail to the user's nail using UV-cured adhesive, then couple the piezo plate at its four corners and center using a 3D-printed spacer and UV glue, similar to common nail-salon techniques. A cross-section of this stackup, excluding the finger, is shown in Figure 3A.

Strong coupling is essential for stability: once cured, the device maintains continuous contact with the body, allowing only bending without sliding or intermittent contact. In this configuration, fingernail bending transfers directly to the piezo plate and is sensed through the piezoelectric effect (Figure 3A). Because piezoelectric materials are bipolar, we can measure both upward and downward strain. This enables detection of simple one-degree-of-freedom pressing gestures as well as differential strains across the four corners of the nail for spatially localized detection. Such patterns support recognition of more complex, multi-degree-of-freedom micro-gestures. These signals are acquired by separating electrodes on the surface of the piezo, as detailed later in this work.

Conversely, applying a voltage uses the reverse piezoelectric effect to bend the plate, transmitting strain through the spacer and artificial nail. The bipolar nature of the material enables multiple actuation modes: upward or downward bending, as well as localized left- or right-side bending using separate electrodes. Adhesives and polymer layers with stiffnesses comparable to commercial artificial nails ensure strong coupling while preserving comfort. This semi-rigid interface efficiently transfers strain yet remains consistent with materials already accepted in everyday cosmetic use.

4 Implementation

Our NailBender prototype consists of three core parts: a piezo bender, a 3D-printed spacer, and an acrylic fake nail (Figure 3C). All of these parts are modified by hand and held together using various adhesives.

4.1 Piezoelectric Actuator

The piezoelectric actuator was one of the most difficult parts to implement, as piezos naturally have very low strain (deformations of only 1-10 μm are common). However, from prior work, we deduced that our piezos would need to produce deflection on the order of 20-30 um between the center and edge of the fingernail, which is typically seen under strong pressing forces [59]. We tried a range of piezo benders with deflection in this range (Appendix Figure 17), and ultimately selected a double-sided piezo plate (13 mm $\times$ 13 mm $\times$ 0.4 mm, Figure 3D) due to its low cost, high force, and low profile.

This piezo has oppositely polarized lead zirconate titanate (PZT) ceramic layers on either side of a flexible brass substrate, which allows for directional bending to occur when a single voltage is applied to both top and bottom electrodes (Figure 3B). We purchased the piezos online from XHXDZ in China (13MM), qty 100 for $5.60. The middle electrode is brass, while the top and bottom electrodes appear silver-based. Based on its flexibility, the PZT appears to be a soft composition, likely a PZT-5 variant. In testing, this piezo

provided the best tradeoff between strength and flexibility. We fabricated over 250 prototypes before final testing and user studies.

We modified the core piezos in two primary ways: patterning the thin silver external electrodes and attaching custom flexible wiring. Patterning the piezo effectively creates separate regions for sensing and actuation, turning one piezo into many. The patterns were designed to maximize active electrode area, since both the sensing charge (Q) and haptic output force (F) scale with area. We therefore favored large, continuous electrode regions over denser patterns to preserve sensing SNR and actuation strength. Patterning was performed by selectively removing surface electrode regions using a handheld drill. This helped reduce damage to the ceramic layer. For wiring, we either soldered each region individually or bridged the top and bottom electrodes. We limited heat on the piezo, as piezoceramics can easily depolarize above their Curie temperature (typically 320–370 °C for soft PZT). This process was faster and more reliable than silver epoxy or conductive copper tape.

4.2 Spacer and Fake Nail

Our 3D-printed spacer connects the piezo layer and the fake nail. The spacer was printed on a Bambu Labs X1C using PLA filament and acts as a planarizing interface between the curved fake nail and the flat piezo disk. In our prototype, the spacer thickness increases from the center apex to approximately 2.5 mm at the lateral edges to accommodate the nail's transverse curvature. In the future, curved piezos could eliminate the need for this spacer and further reduce the stackup. After cleaning the piezo and spacer surfaces with IPA, the piezo was bonded to the spacer using a medium-viscosity instant adhesive (3M PR1500). We used commercially available ABS fake nails in standard multi-size sets and selected the best-fitting nail for each user to accommodate differences in nail geometry. The fake nail was mounted to the user's natural nail using solid UV-curable gel stickers and cured under UV light for 2–5 minutes. The piezo-spacer assembly was then attached to the fake nail using UV-based glue to better conform to the user's nail shape. For removal, we used a non-acetone-based remover and a wooden stick to separate the fake nail from the natural nail. This process closely follows common nail art procedures, making it easy to replicate.

4.3 Electronics

We developed a custom, multi-channel (4 in × 4 out) set of miniature, lightweight (5 × 2 × 2 cm, 16 g) electronics for our NailBender prototypes, user study, and application scenarios (Figure 4). They consist of modular sensing and actuation circuits, which connect to the piezo and interface with a Teensy 4.0 micro-controller through GPIO and ADC pins in a stacked configuration. The Teensy communicates with a PC over USB, and the board is powered by a 5 V USB supply. We implemented custom 5 V to +/- 90 V DC/DC boost converters based on the LT3484 ($7.78) monolithic controller and two miniature flyback transformers ($1.30 TDK ATB322515). Sensing electronics were powered from the 3.3 V LDO on the Teensy.

A simplified schematic of a single channel of actuation and sensing (Figure 4C). For actuation, we used the HV2621 ($27.53) high voltage, 16-channel multiplexer to selectively connect the piezo actuation electrode to +90 V, -90 V, or ground. The sensing electronics were built around the ADA4505-4 ($1.64), a low-cost, 4-channel

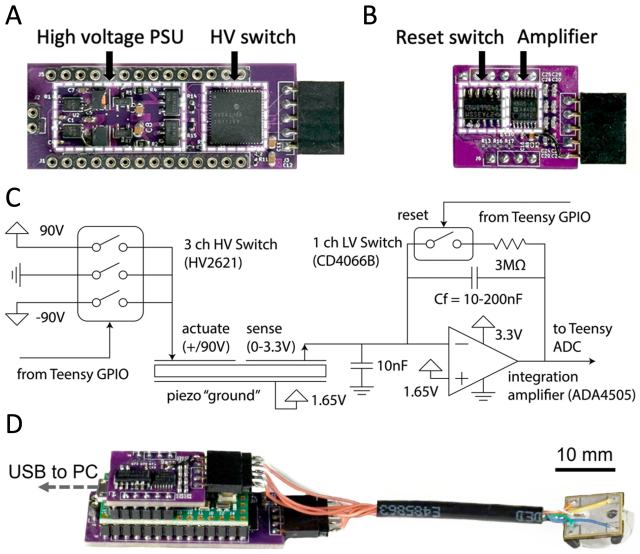

Figure 4: A) Custom high-voltage supply and switch array. B) Low-voltage sensing amplifier and reset circuit. C) Simplified schematic of one channel in the four-channel device. D) Fully assembled electronics connected to the NailBender via USB.

op-amp with exceptionally low input current (0.5 pA), and supply current (10 μA per channel). We put this op-amp as an inverting integrator to convert the charge generated by the piezo ($F \propto Q$) into a steady voltage. Piezos, however, are notoriously difficult to compensate for drift (as little as 10 pA can be disastrous), and so we also included a reset switch, coupled through a 3 MΩ resistor, using a commodity 4 channel CMOS switch CD4066B ($0.52). The system is referenced to 1.65 V in order to avoid needing a negative voltage rail. We cleaned the electrodes with IPA and ensured good insulation to minimize sense-line leakage. To reduce mechanical noise, all four channels were routed through a single multi-conductor cable, which reduced bending-induced artifacts. In practice, this sensor drift remained a small, nagging consideration, though it was mostly mitigated with our reset switch and careful cleaning. The feedback capacitor C_f sets the main gain of the device, and we optimized it between 10-200 nF depending on the surface area of the sensing piezo used, as a larger surface area produces more charge and requires less gain. This optimization tunes the sensitivity to suppress the response to incidental contact, ensuring that the stronger signals from intentional micro-gestures remain clearly distinguishable. The charge amplifier output was sampled by the Teensy ADC and streamed to a Python PC application at an effective throughput of 3,500 Hz. We implemented four actuation and four sensing channels, though only a subset was used in practice.

We briefly measured the overall power consumption of the electronics from the USB supply over a 5-minute average. The idle system consumed 0.39 W, dominated by the microcontroller, while enabling the sensing circuit produced no detectable increase in power, consistent with the amplifiers' 40 μA current draw. Driving a full NailBender device at 20 Hz increased input power by only 40 mW, highlighting the system's low-power operation.

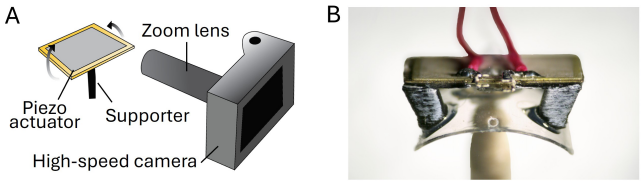

Figure 5: A) Experimental setup for physical motion characterization. B) Camera image of our NailBender grounded in the center with motion tracked at the edges.

4.4 Safety

The primary electrical safety concern with NailBender is its high-voltage piezo drive. Although the ±90 V_{pk} drive used in our characterization exceeds SELV voltage limits, the user does not directly contact the high-voltage circuitry, and we mitigated risk through multiple safeguards, including low-power custom DC/DC converters, inline current limiting to the milliampere range, isolating switches, and polymer insulation on exposed outputs. Additional risks, including UV glue exposure, minor nail injury, and lead in PZT, were minimized through cosmetic-grade materials, controlled preparation, and the chemical stability of PZT under normal use.

5 Single input/output configuration

We first describe the operation of a single-input, single-output NailBender device, including its raw sensing and actuation capabilities.

5.1 Actuator Performance

To characterize the piezo actuator and assembled NailBender system, we used a high-speed camera (Figure 5A) with microscopic zoom lens (Kron Tech Chronos HD 2.1) and motion tracking software. The center of the piezo was mechanically grounded, and the deformation at the actuator edge was recorded relative to center (Figure 5B). The results show that the selected piezo exhibits an omnidirectional radial bending mode, as expected, producing a free displacement of approximately 140 μm across the operating frequency range of 0–100 Hz under an applied voltage of ±90 V_{pk} or 180 V_{pp} (Figure 6A). Notably, these piezos are nominally rated for actuation below 30 V, likely due to the risk of ceramic fracture at

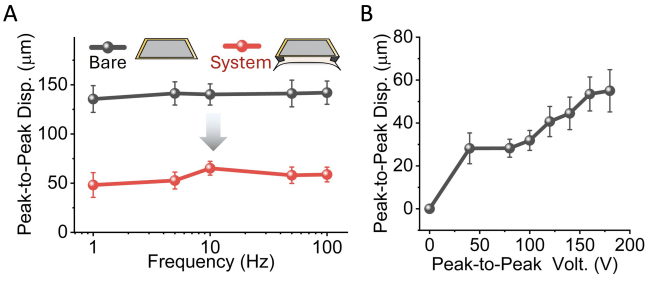

Figure 6: A) Motion displacement results for a bare piezo, and the full NailBender system. B) Amplitude shown as a function of applied voltage.

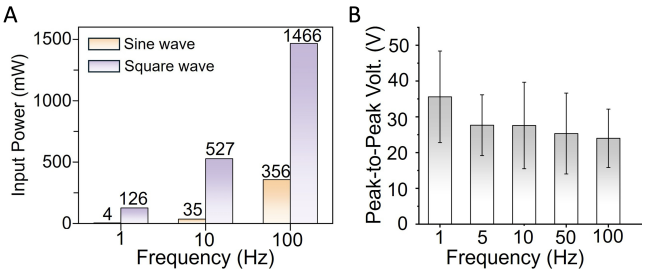

Figure 7: A) Piezo input power under different waveforms. B) Psychophysical absolute detection thresholds.

higher deformations, a failure mode we also sometimes observed during testing with unconstrained, bare piezos.

When bonded to our spacer and fake nail, the maximum displacement was attenuated by about half, to under 50 µm (Figure 6A). We believe this allowed the application of higher voltages (up to ±90 V). We confirmed that operation at these voltages did not induce a strain–electric field hysteresis loop [13], a phenomenon we had observed in other piezos during early prototyping. This non-linearity can be common when driving piezos to their extremes. Actuator displacement scaled linearly with input voltage (Figure 6B). We also measured the power consumption of the piezo at 1, 10 and 100 Hz under both sinusoidal and square-wave input voltages with an amplitude of ±90 V. As shown in Figure 7A, input power increased with frequency, as expected for capacitive loads without energy recovery. Sine-wave drive remained below approximately 0.36 W, while square-wave drive reached 1.47 W at 100 Hz, though this power was largely reactive. No noticeable heat generation was observed during testing.

5.2 User Study I: Detection Thresholds

To evaluate the base effectiveness of NailBender, we measured absolute detection thresholds (ADTs) for input voltage under square-wave stimulation at 1, 5, 10, 50, and 100 Hz. We utilized a dual ascending and descending transformed 1-up–2-down double-staircase procedure to estimate the 70.7% detection threshold on the psychometric function. Ten participants (2 female, 8 male) were recruited for the experiment. This study was approved by the University of Illinois Urbana-Champaign IRB, and all participants across our studies provided written informed consent. Participants wore a headset playing continuous white noise and responded "Yes" or "No" in each trial depending on whether they perceived the stimulus. After eight staircase reversals, the absolute detection threshold was calculated as the mean of the last four reversals. As shown in Figure 7B, low-frequency actuation yielded the highest detection thresholds. Importantly, all participants were able to perceive the stimulus at all tested frequencies with an input voltage below 40 Vpp. These results demonstrate that NailBender can deliver perceptible haptic feedback at low voltage thresholds and with low actuation power.

5.3 Sensing Performance

In Section 3.1, we noted that pressure at the fingerpad can be estimated by monitoring strain in the fingernails. One advantage of

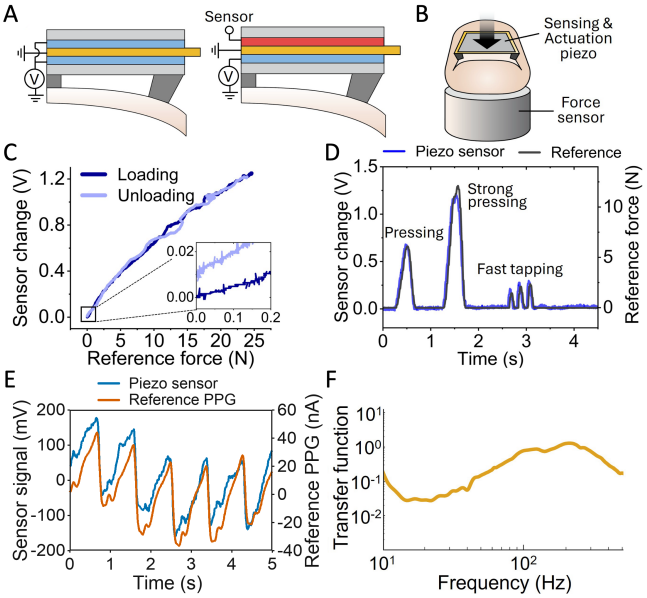

Figure 8: A) Actuation-only and sensing & actuation test configuration. B) Force measurement setup. C) Sensor output versus reference force. D) Response to pressing, tapping, and strong presses. E) Heartbeat with reference PPG (no load). F) Frequency response to broadband vibration.

piezoelectric actuators is that they provide both actuation (via the inverse piezoelectric effect) and self-sensing (via the direct piezoelectric effect) in a single device. However, when we attempted to use a single set of electrodes for both functions, we observed significant interference that degraded sensing reliability. The main reason is that the output drivers have too much charge leakage, even when turned off, making it difficult to fully isolate the actuation driver from the sensing electronics.

To address this limitation, we physically separated sensing and actuation within the piezo. We dedicated the top layer to sensing (Figure 8A) and the bottom layer to actuation in this section, unlike the previous section where both top and bottom layers were used for actuation. We also coupled the piezo brass, a common electrode shared by sensing and actuation, to a 1.65 V reference through a 10 uF capacitor. This enabled stable concurrent actuation and sensing.

To characterize one-degree-of-freedom (1-DOF) force sensing, we compared the piezo sensor output with reference measurements from an ATI Nano17 force sensor (Figure 8B). The sensor reliably detected forces over a wide range, from light touch (0.1 N) to strong pressing (24 N) (Figure 8C). We observed hysteresis and drift in the sensor output, likely due to biomechanical factors of the finger, including skin compliance, subcutaneous tissue, and nail-bed deformation. Time-series data further showed rapid capture of fingertip force dynamics (Figure 8D). The sensor could also detect fingernail heartbeat signals that tracked a reference PPG (MAX30102), suggesting potential for wearable biosensing (Figure 8E). Finally, we characterized bandwidth by placing a finger on an electromagnetic shaker and delivering a 10-500 Hz chirp to the fingerpad, confirming measurable responses across this range (Figure 8F).

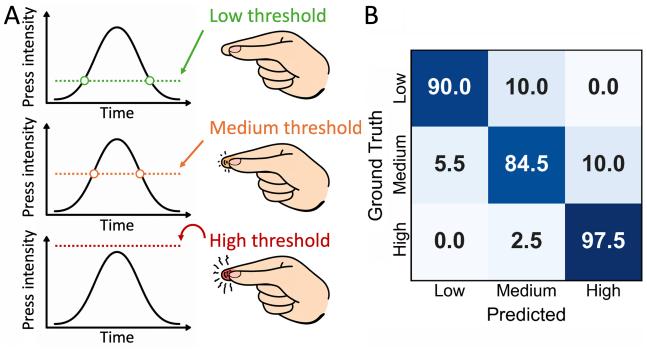

Figure 9: A) Three button press configurations. B) Confusion matrix for button press thresholds.

5.4 User study II: Variable-Force Pinch Button

To validate combined tactile sensing and feedback, we conducted a user study that rendered button clicks at different pressure thresholds. We monitored force with our sensing electronics, and triggered haptic feedback at predefined thresholds. First, participants lightly pinched their index finger against the thumb, and the sensing signal level was recorded. Next, participants generated a maximum-effort pinch, and the corresponding sensing signal was measured. Using these two reference points, we defined three thresholds for triggering feedback: low, medium (30% of the full range), and high (Figure 9A). Once the button value exceeded these thresholds, a haptic pulse (20 Hz for 200 ms) was applied.

After a two-minute training session, each threshold condition was repeated 20 times in random order, resulting in a total of 60 trials (3×20). Participants were encouraged to vary their pressing force to explore the triggering thresholds. After each trial, participants reported their perceived tactile button configuration. The overall accuracy of the test was 90.67% ± 4.66% (Figure 9B). The high threshold achieved 97.5% accuracy, with most misclassifications occurring between the low and medium thresholds.

Participant feedback suggested these misclassifications were influenced by several factors. One participant noted that press speed affected perception, with repeated presses making force differences easier to distinguish than presses separated by a few seconds. Another reported that press location affected triggering, with lighter touches more likely to activate feedback at the fingertip than further back on the finger. Despite confusion, three participants preferred the medium-force condition, finding it realistic without being too tiring. Overall, this study shows that NailBender can detect pressing force and provide real-time tactile feedback, supporting interactions in both virtual and real environments.

6 Multi-output Configuration

To investigate a multi-output configuration we segmented our piezo disc into two regions, filing lines down the middle of both top and bottom sides and connecting them with bridging wires (Figure 10A).

6.1 Segmented actuation

To validate that the segmented piezo disc provides spatially differentiated feedback, we first compared edge displacement between

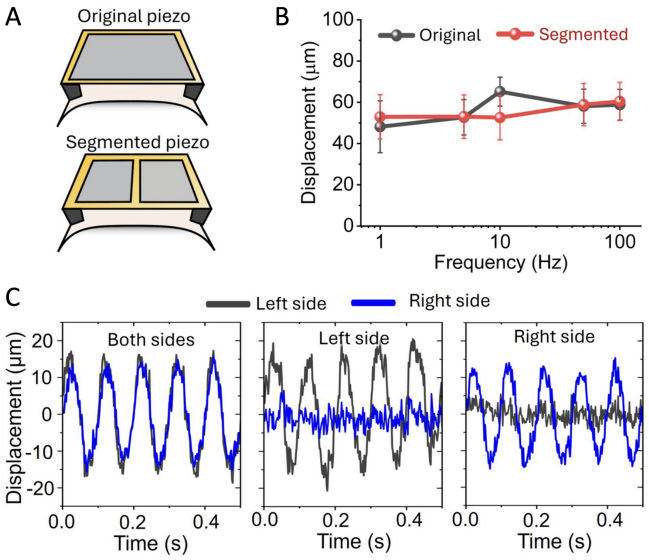

Figure 10: A) Piezo segmentation for multi-output configuration. B) Displacement comparison of segmented piezo compared to non-segmented piezo. C) Time domain recording showing good motion separation between left and right sides of segmented piezo.

the single and segmented actuators under a square-wave input voltage of ±90 V. Results showed negligible performance degradation due to electrode segmentation (Figure 10B). We then measured the displacement at each actuator edge when actuating both sides together, only the left side, and only the right side. Results showed that the segmented piezo disc produces spatially differentiated displacement on each side through independent voltage control, with little crosstalk. However, since the rigid piezo-nail assembly efficiently propagates vibrations across the interface, we believe that finer segmentation would offer minimal gain in spatial resolution due to mechanical coupling.

6.2 User study III: Pattern recognition

Building on our characterization of localized deformation in the segmented piezo disc, we conducted a user study to examine whether NailBender could convey spatially differentiated tactile patterns. We designed five patterns: simultaneous activation of both sides, left only, right only, sequential left-to-right, and sequential right-to-left, with a 0.4 s delay between activations. Each pattern used a 20 Hz vibration lasting 2 s. These conditions were chosen to evaluate NailBender's ability to render both spatially differentiated feedback and dynamic directional transitions across the nail surface.

The experiment involved the same ten participants recruited previously. To eliminate auditory cues, participants wore headphones playing white noise throughout the trials. Before the formal test, they completed a 5-minute training session. During the test, each pattern was presented 20 times in randomized order, yielding a total of 100 trials (5 patterns × 20 repetitions). The results show that participants were generally able to distinguish five different patterns, achieving an overall accuracy of 83.2% (Figure 11).

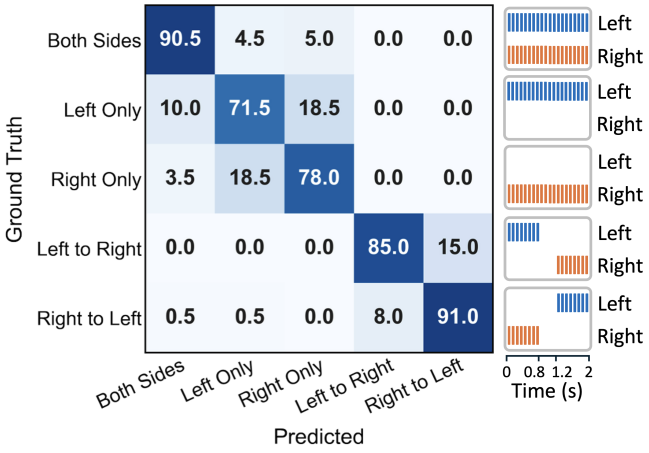

Figure 11: Spatially differentiated pattern recognition results.

The main confusions arose between the 'Left Only' and 'Right Only' conditions, suggesting that distinguishing single-sided feedback is more challenging. We attribute this to mechanical coupling between the rigid actuator and the nail plate. Although the actuator itself produces localized vibration (Figure 10C), the assembled structure propagates these vibrations more globally. This blurring of spatial distinctions reduces the perceptual salience of lateral differences, suggesting that finer electrode segmentation may provide limited perceptual benefit. This limitation may be mitigated by using static (or very low-frequency) forces instead of vibration, although this would require further improvement of the actuator's output force and remains future work.

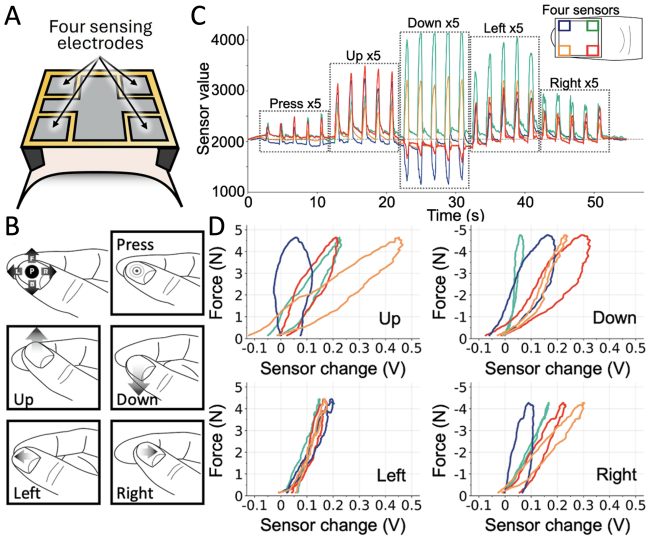

Figure 12: A) Electrode configuration. B) Gesture labels for micro-gesture testing. C) Raw sensor data showing distinct, repeatable trends across gestures. D) Sensor output versus reference force for four micro-gesture directions.

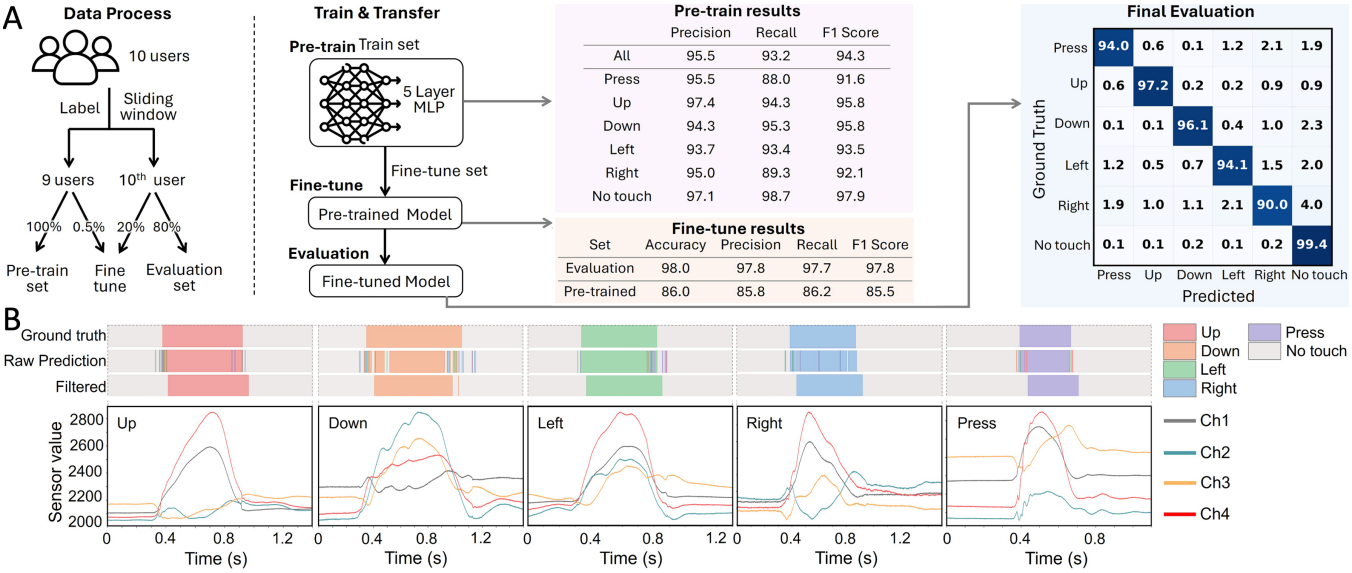

Figure 13: A) ML pipeline overview: data collection, training and summary results, and confusion matrix. B) Example final-evaluation results showing raw signals, ground truth, raw predictions, and 30-sample majority-vote smoothed predictions.

7 Multi-Input Configuration

In our final configuration, we wanted to see if we could detect multiple strain inputs with the goal of sensing more complex tactile events than a simple pressing (Section 5). To achieve this, we adopted a four-quadrant sensing layout. This configuration provides sufficient spatial resolution to capture the main axes of finger deformation (e.g., pitch, roll, and press) while preserving the central area for haptic actuation.

We separated the top layer into five electrodes: four sensing electrodes at the corners and one center electrode that can be bridged with the bottom electrode for actuation (Figure 12A). We chose the corner layout to capture strain between the piezo center and the four mounting locations. Due to the smaller sensing area, we reduced the feedback capacitor to 10 nF to increase amplifier gain.

Once mounted to a finger, the device produced distinct signal patterns for the micro-gestures shown in Figure 12B. All sensors exhibited an initial loading response as the thumb pressed into the index finger, followed by rapid changes during micro-movements. After unloading, we observed a brief peak and characteristic decay, with similar hysteresis and baseline shifts also visible in the force-sensor curves (Figure 12D). These effects likely arise from finger-pad viscoelasticity, a well-known biomechanical response to indentation [54]. They were less pronounced in the single-input configuration (Figure 8). We further confirmed that they did not originate from the piezo or amplifier by applying constant strain and observing a stable output. Despite these effects, the signals remained generally repeatable across gestures, as illustrated by five examples per gesture in Figure 12C.

Our five gestures, press and in-plane up, down, left, and right (Figure 12B), were designed as force-based inputs without intended slip, although eliminating slip entirely in practice was difficult. We observed strong trends for press and up/down gestures, but weaker trends for left/right, with additional variation across users and devices likely due to differences in finger biomechanics.

Based on these observations of clear gesture-dependent trends, together with hysteresis and baseline shifts, we hypothesized that a per-user calibrated machine learning (ML) approach could classify these gestures. We adopted a transfer-learning approach, pre-training across users, then fine-tuning on a small per-user calibration set, and evaluate it both offline and in simulated real-time. An overview of our approach and summary of results is in Figure 13A.

7.1 Data Processing and Training

We recruited 10 participants. After a brief familiarization phase, each performed five gestures plus a no-touch condition, one two-minute trial per condition (60 trials total). Trials used written on-screen instructions, and individual gestures lasted about 0.5 s. In total we collected approximately 100.8 M data points (10 users × 6 conditions × 120 s at 3,500 Hz × 4 sensor channels).

After data collection, we labeled the periods between gestures as "no-touch" using a custom script followed by manual adjustment, providing ground-truth labels for training. We then used 10-sample non-overlapping windows to reduce the prediction rate to 350 Hz, or one prediction every 2.86 ms. The final training input consisted of 4 channels × 10 samples, for a total of 40 dimensions.

We implemented a lightweight five-layer MLP with ReLU activations, trained using cross-entropy loss for low-latency inference. Its architecture and hyperparameters were selected from held-out training-user data and fixed across users. The model was trained on gesture data from 9 users, with the 10th user held out for evaluation. To adapt to new users, we pre-trained on 80% of the training data and fine-tuned on a small calibration set from the target user, requiring only two minutes of data collection and training. Final evaluation on the remaining held-out data estimated real-world accuracy.

7.2 Micro-Gesture Results

Results were computed in scikit-learn using accuracy, precision, recall, and F1-score. For pre-training evaluation, 20% of the training set was held out before training. We report per-class results for the pre-trained model, overall fine-tuning performance on both sets, and the evaluation-set confusion matrix (Figure 13A).

The pre-trained MLP achieved 96.41% overall accuracy. After fine-tuning with part of the 10th user's data, it retained 86% accuracy on the original pre-training evaluation set and achieved 98% on the evaluation set, demonstrating strong personalization from a small adaptation set. This 98% reflects a per-user-calibrated setting rather than fully user-independent recognition. By comparison, training from scratch on the same fine-tuning subset (0.5% of the training set + 20% of the 10th user's data) achieved only 75.17% accuracy, highlighting the benefit of pre-training when adaptation data are limited. Thus, the two-minute calibration is not intended to train a complete model from scratch, but to adapt a pre-trained recognizer to the signal characteristics of a new user.

To qualitatively assess ML performance, Figure 13B shows exemplar test-set predictions for each gesture class. Raw sensor data are shown with ground-truth labels and fine-tuned model predictions. Most remaining errors stemmed from gesture transitions and transient artifacts. We reduced this residual noise with a 30-prediction (85 ms) majority vote filter, which stabilized transitions and rejected short spurious classifications from environmental interference with only a small latency penalty. The filtered predictions were then used to trigger the final gestures. Finally, our results are broadly comparable to prior work using multiple strain gauges (10 written digits + 5 broad tasks, or 10 classes [59]). This suggests that the informative signal arises from spatial fingernail flexion rather than an idiosyncrasy of our sensor, and that a piezo-based approach may achieve similar performance to strain-gauge methods, although direct comparison is beyond the scope of this work.

8 Application Scenarios

Prior work has motivated haptically transparent, fingerpad-free haptic interfaces [71, 74–76, 79, 83]. Rather than reviewing this space comprehensively, we highlight several scenarios to contextualize NailBender and illustrate its input-output capabilities.

8.1 Real World Guidance

A NailBender device configured with left and right output actuation and a single input sensor (similar to the device from Section 6) can be used to provide feedback in real world situations where user's hands are pre-occupied and their vision is focused elsewhere, such as when riding a bike. In this situation, the user cannot access the phone in their pocket, and checking a smartwatch or using earbuds could be too distracting or isolating. Instead, a NailBender device with main electronics mounted on the wrist (Figure 14A) communicates to their smartphone, which is running a GPS guidance application.

When approaching a stop sign (Figure 14B), the user receives a buzz on their fingertip reminding them to brake and watch for traffic. They can press their fingertip against the handlebar or brake lever to dismiss it while maintaining full control of the bike (Figure 14C). Once stopped at the intersection (Figure 14D), the user receives clicks on the right side of the finger, like a car blinker, indicating a right turn (Figure 14E). They can again press the handlebar to dismiss this, or the GPS app stops clicking once progress is detected. This scenario is presented as a design example rather than a formally evaluated system; riding introduces motion and vibration that could affect perception, and the left/right confusion we observed would likely be more pronounced in this setting.

8.2 Mixed Reality Control

In this scenario, a user wears a NailBender device that can detect multi-directional input and a single output (Section 7). This device is always attached to their finger and communicates to a mixed reality headset. As they are sitting on the couch at home, they discretely pinch their fingers at their side. This brings up a UI menu in front of them (Figure 15A-D). They navigate up and down through this menu using micro-gesture force pinch inputs, with the device configured to provide nuanced feedback (20 Hz, 0.4 s) for navigation and distinct tactile clicks (10 Hz, 0.2 s) to confirm selections. They force-press on their photos app (Figure 15D-H), opening a gallery on their mixed-reality photo frame of photos recently sent by a friend. After gesturing left and right to swipe between photos (Figure 15E-G), they gesture upward to "like" one. Finally, they double force-press to close the AR menu. This scenario highlights real-time closed-loop interaction with concurrent sensing and haptic output.

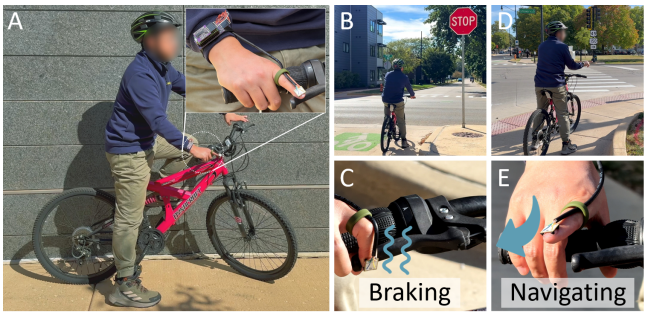

Figure 14: A) User wearing a multi-output NailBender while biking. B) Approaching a stop sign and C) receiving a buzz to brake. D) Turning right after E) right-side taps on the finger.

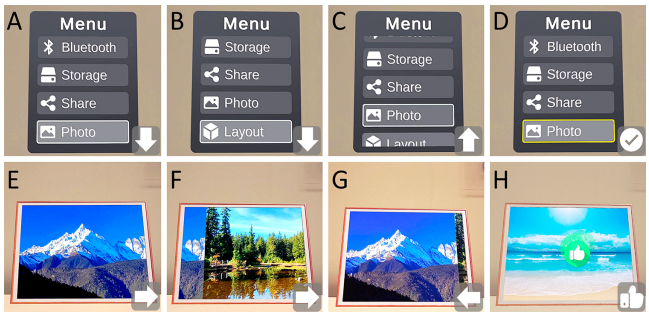

Figure 15: A-D) Menu navigation with NailBender micro-gestures. E-H) Swiping through photos on an MR photo frame and liking a friend's photo.

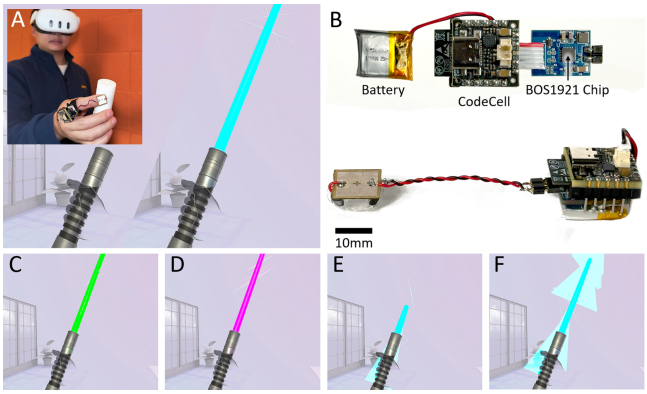

Figure 16: A) User in VR with a lightweight, wireless, battery-powered NailBender and an unactuated lightsaber prop. B) Ultra-small wireless prototype compared with NailBender. C-F) Button presses change lightsaber color, trigger extension/retraction, and provide vibrotactile nail feedback.

8.3 VR Gaming

In our final scenario, we developed an extra-compact ring-based electronics package (20 × 17 × 24 mm) built around the Boreas BOS1921 piezo driver, enabling both sensing and actuation in a single ultra-low-power package with capacitive energy recovery (Figure 16). This design could be further integrated into a small magnetic "finger-bud" that clips directly onto NailBender. In VR, the user picks up an unactuated prop affording a closed-hand grip; the grip is detected and a virtual lightsaber is rendered. Pressing the side of the prop activates the saber, felt as a gentle hum through the fingernail, and further presses cycle the beam color, each with a distinct vibration, while a double tap turns it off.

9 Limitations and Future Work

This work demonstrates a compact fingernail-mounted piezo interface for both sensing and actuation, but several limitations remain.

Low-frequency haptic perception was limited by the piezo's small actuation output. Some participants in User Study I could not perceive a 1 Hz sine-wave stimulus, indicating that piezo displacement was still insufficient. In this study, we used a double-stacked piezo plate, but additional layers could further increase output in future designs. Spatial resolution was similarly constrained: User Study III suggested that nail rigidity reduced perception of localized feedback, with left-only and right-only stimuli being the most frequently confused pair. Mechanical decoupling between actuator regions, such as introducing more compliant intermediate layers, could be explored in future work to reduce cross-nail propagation and improve lateral resolution.

Long-term stability remains a challenge, as piezo elements are prone to cracking under external loads. We frequently observed NailBender damage during removal, and repeated wear or prolonged use further increased failure risk. Softer artificial nail materials could reduce removal stress. We also considered softer PVDF-based piezo films, but their lower piezoelectric output makes strong actuation and sensitive sensing more difficult.

Our current prototype is lightweight (0.84 g) and compact, but future designs could be miniaturized further by mounting a curved piezo bender directly on the nail. Customization remains important because nail size and shape vary across users, and smaller nail plates intensify the sensing actuation trade-off. Future work should explore flexible attachment strategies and long-term comfort.

Another limitation is the driving voltage. The current implementation requires relatively high voltages (±90 V) for sufficient actuation, complicating integration into compact wearables and limiting future multi-finger or mobile use. Lowering the required voltage would improve efficiency, safety, and practicality.

The reported 98% micro-gesture accuracy reflects a per-user-calibrated setting. Because NailBender is worn as a fixed artificial nail over extended periods rather than repeatedly removed and remounted, per-user fine-tuning is performed only once, and the dataset spans multiple NailBender instances, giving some exposure to device-to-device variation. While the reset switch mitigates sensor drift during use, this was a small-scale study, and we did not run a dedicated evaluation isolating re-mounting, long-term drift, or repeated wear, which remains future work.

While discrete micro-gestures enabled effective real-time interaction, continuous micro-gestures remain more challenging. User Study II demonstrated real-time pressure sensing with a single piezo sensor, and multi-input sensing may further enable estimation of gesture direction and magnitude. The variable-force pinch button also relied on threshold-based sensing, which we found sensitive to press speed and contact location. Adaptive thresholding could be explored to improve robustness in future work. However, robustness to real-world interference, such as unintentional movements, and to actuation-related interference remain important challenges for closed-loop interaction. In this work, actuation interference was mitigated using a brief post-actuation sensing blanking period (<1 s). Larger multi-user datasets may help improve generalization and better distinguish intentional inputs from environmental noise [25], supporting more robust continuous interactions. Finally, all three studies shared the same ten participants, and evaluating across a more diverse user population remains important future work.

10 Conclusion

We presented NailBender, a nail-mounted interface platform that treats nail flexion as a shared channel for both sensing and actuation. Using a single piezoelectric plate with segmented electrodes, our device leaves the fingerpad unobstructed while supporting multiple input channels for force and micro-gesture sensing, and multiple output channels for spatially differentiated and distinct haptic feedback. Our technical characterization and user studies show that this minimal hardware can enable expressive interactions and illustrative applications such as everyday guidance, MR control, and gaming.

Looking ahead, we see opportunities to reduce drive voltage, improve durability and comfort, expand to multi-finger configurations, and explore continuous force and pose control beyond discrete gestures. We hope this work encourages further exploration of nail flexion as a compact, wearable substrate for richer multi-input and multi-output haptic interaction.

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A Supplementary Actuator Details

The actuators explored during early prototyping (Appendix Figure 17, shown left to right) included a small PZT-5 element (5 mm × 3 mm × 0.2 mm), a thin piezo bender (15 mm × 1.5 mm × 0.6 mm, StemInc SMBS1515T6P750WL), a stacked piezo (7 mm × 3.75 mm × 1.22 mm, TDK PowerHap 0704H013V060), and three piezo discs (7.6 mm diameter × 0.17 mm, PUI AB0818B; 12 mm diameter × 0.32 mm, PUI AB1290B-LW100-R; and 15 mm diameter × 0.28 mm, PUI HD-PAB1501), before we ultimately selected the thin, double-sided piezo plate used in the final design (13 mm × 13 mm × 0.4 mm, far right), with two oppositely polarized PZT layers.

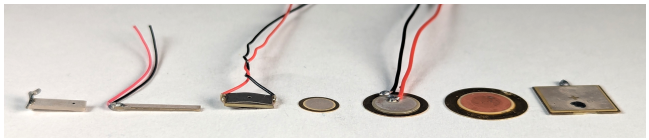

Figure 17: Piezos tried during prototyping, left to right. We chose the double-sided square piezo, far right.